



AWS SCALING PROJECT

Presentation

NOTRE EQUIPE



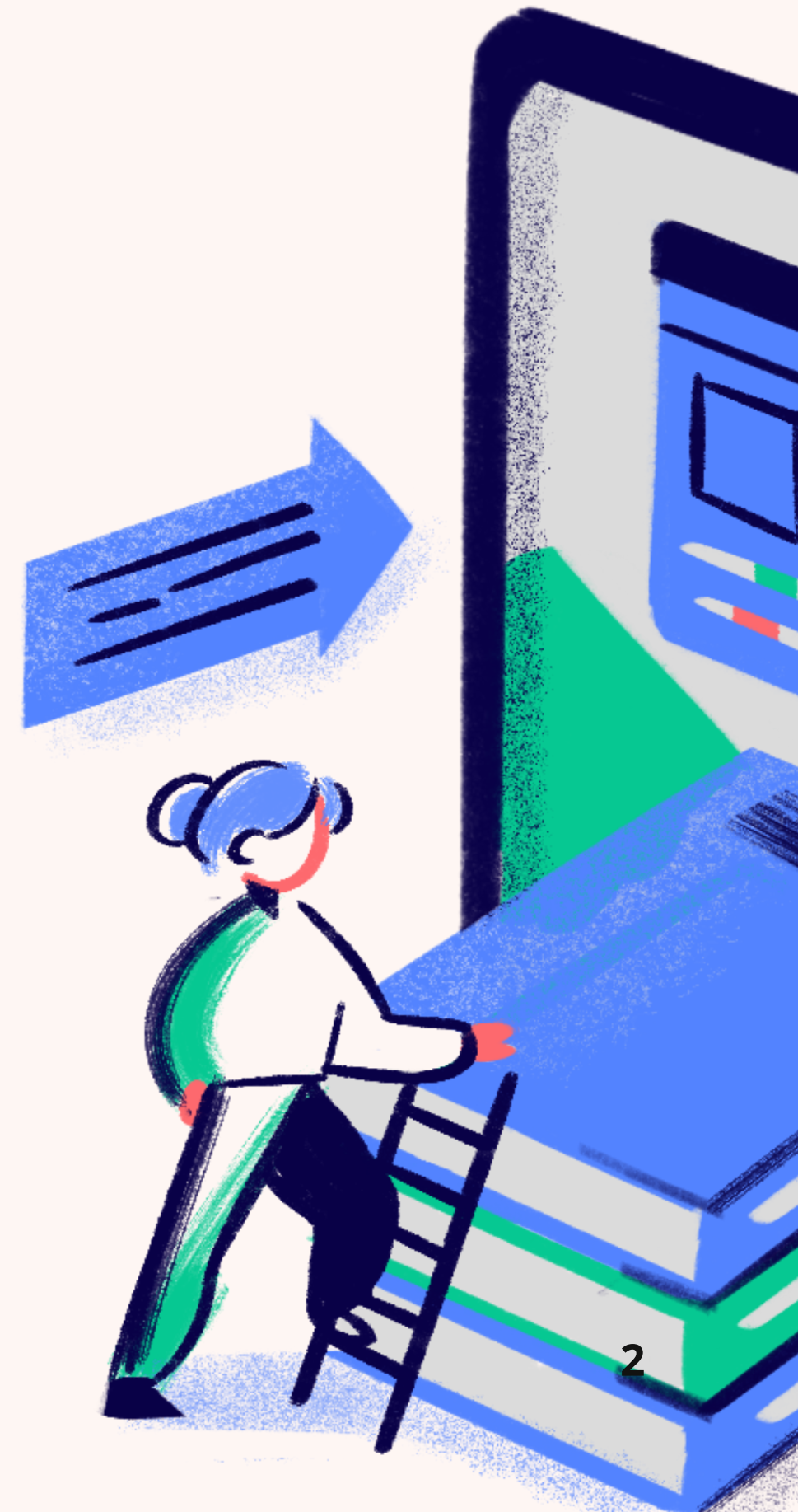
Enzo FERRARI

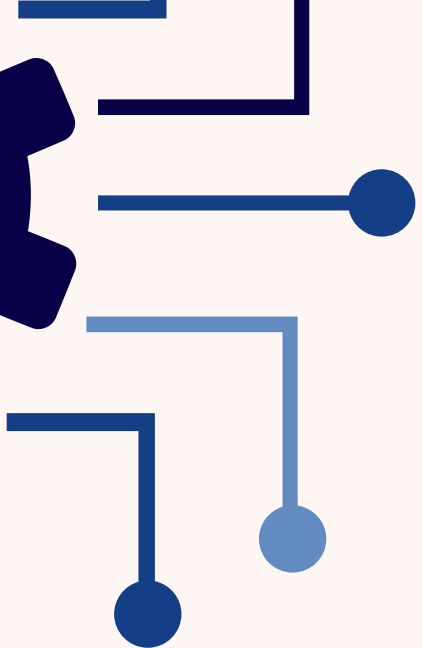


Haunui TEHAHE

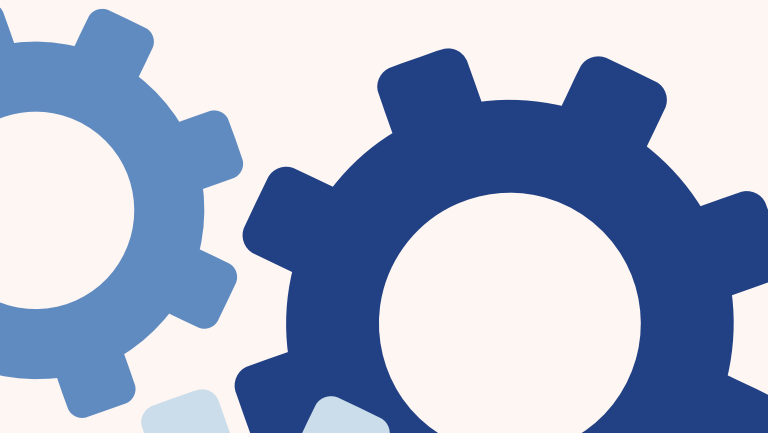


Aurélien BESOMBES



In the top left corner, there are decorative blue lines and dots resembling a circuit or network diagram.

SOMMAIRE

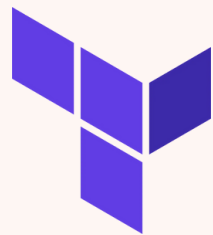
In the bottom left corner, there are two interlocking blue gears of different sizes.

- 01** CONTEXTE / OBJECTIFS
- 02** ORGANISATION
- 03** STRUCTURE
- 04** INFRA
- 05** APP
- 06** MONITORING
- 07** DEMO



CONTEXTE / OBJECTIFS

Déploiement



Observabilité



Tests de déclenchement



Locust

ORGANISATION

aws

aws-project-scaling

View 1 + New view

Filter by keyword or by field

Backlog 0

Todo 0

This item hasn't been started

In Progress 2

This is actively being worked on

Draft
Create presentation slides
Closing

Draft
Prepare demo script
Closing

Done 24

This has been completed

Draft
Create Git repository with folder structure
HIGH

Draft
Write load test scenarios
Monitoring

Draft
Full infrastructure deployment test
Closing

Draft
Create scale-down CloudWatch alarm and policy
MEDIUM Infra 5

STRUCTURE

aws

INFRA

APP

MONITORING

```
terraform
├── main.tf
├── modules
│   ├── autoscaling
│   │   ├── main.tf
│   │   ├── outputs.tf
│   │   └── variables.tf
│   ├── loadbalancer
│   │   ├── main.tf
│   │   ├── outputs.tf
│   │   └── variables.tf
│   ├── monitoring
│   │   ├── main.tf
│   │   ├── outputs.tf
│   │   └── variables.tf
│   └── networking
│       ├── main.tf
│       ├── outputs.tf
│       ├── terraform.tfstate
│       ├── terraform.tfstate.backup
│       └── variables.tf
├── outputs.tf
├── providers.tf
├── terraform.tfstate
├── terraform.tfstate.backup
├── terraform.tfvars
└── variables.tf
```

```
app
├── app.service
├── nginx.conf
├── load-testing
│   └── scenarios
│       ├── baseline.json
│       ├── scale-down.json
│       └── scale-up.json
```

```
monitoring
├── grafana
│   └── dashboards
│       └── infrastructure-overview.json
├── prometheus
│   └── prometheus.yml
├── setup-monitoring.sh
└── README.md
```



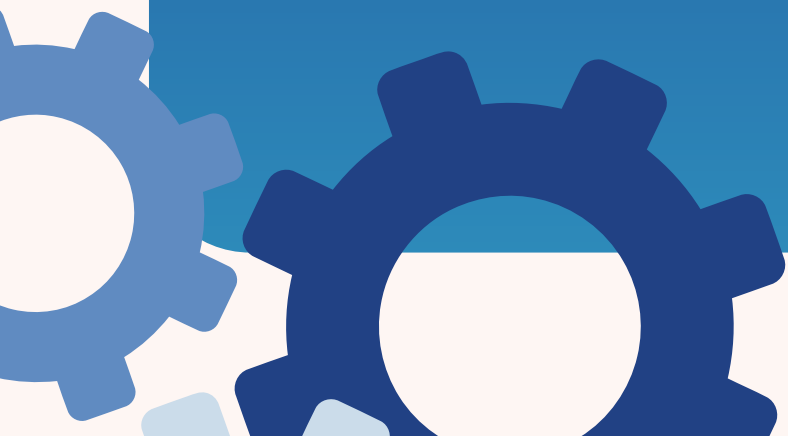
Module Networking

- Création du VPC
- Création des Subnets (Private + Public)
- Création des security group
- Création des tables d'associations



Module Loadbalancer

- Création LoadBalancer public
- Création d'un TargetGroup http
- Configuration du listener HTTP sur le port 80

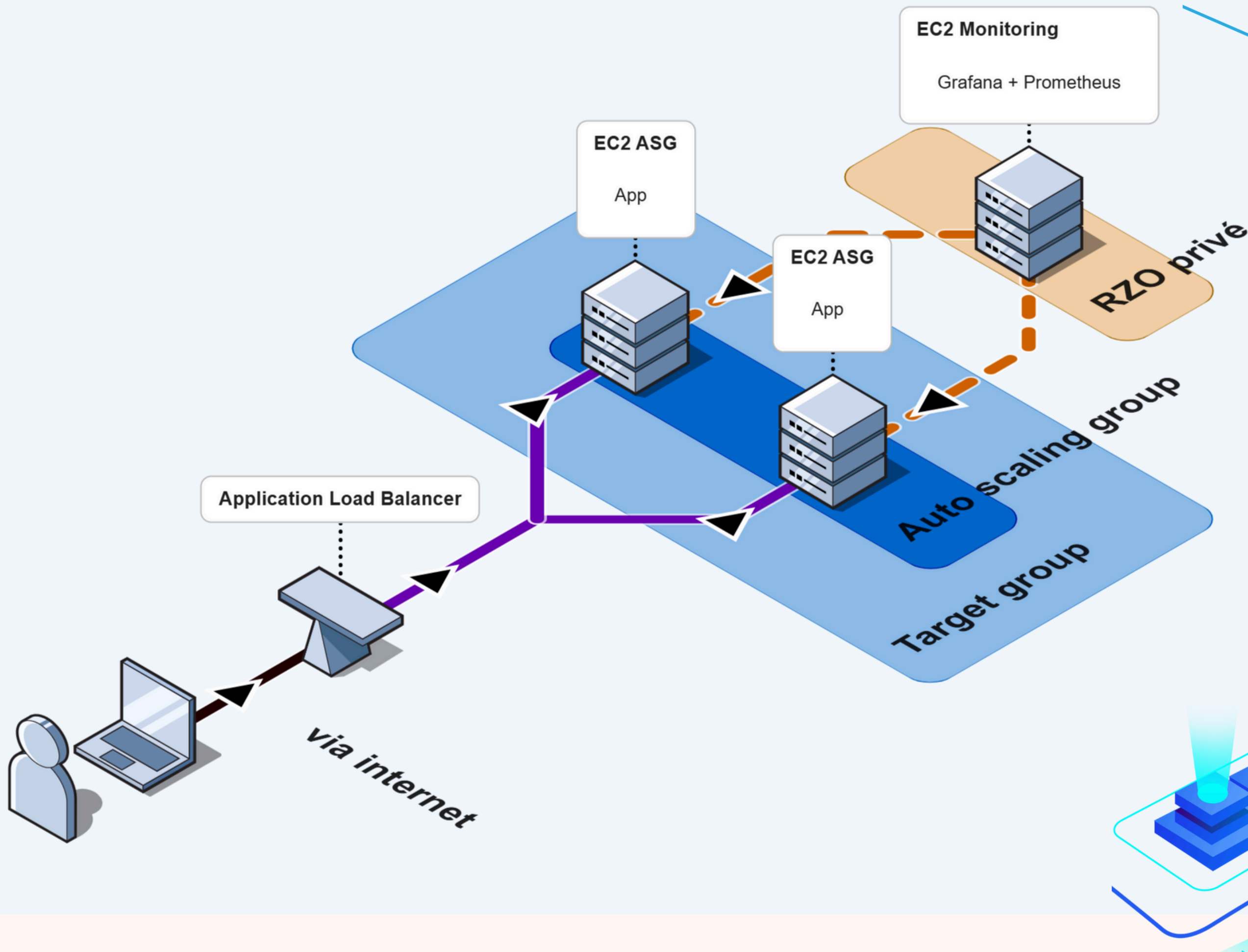


Module Autoscaling

- Définitions du modèle de lancement des instances EC2
- Création des instance EC2 scalable (scaling horizontal)
- Configuration de la partie scaling policies



aws



APP

aws

App accessible via le DNS du load balancer



AWS Auto-scaling Demo



Availability Zone:



Private IP:



Hostname: ip-10-0-1-55



Timestamp: 2026-01-20 12:57:05

MONITORING



Prometheus

Prometheus

Métriques node exporter des EC2
Port 9100

Job ec2-node
tag: environnement = prod
accès autorisé par IAM



Grafana

Grafana

%CPU moyen des EC2
%RAM par EC2

Nombre EC2 actives
Nombre EC2 éteinte

MONITORING



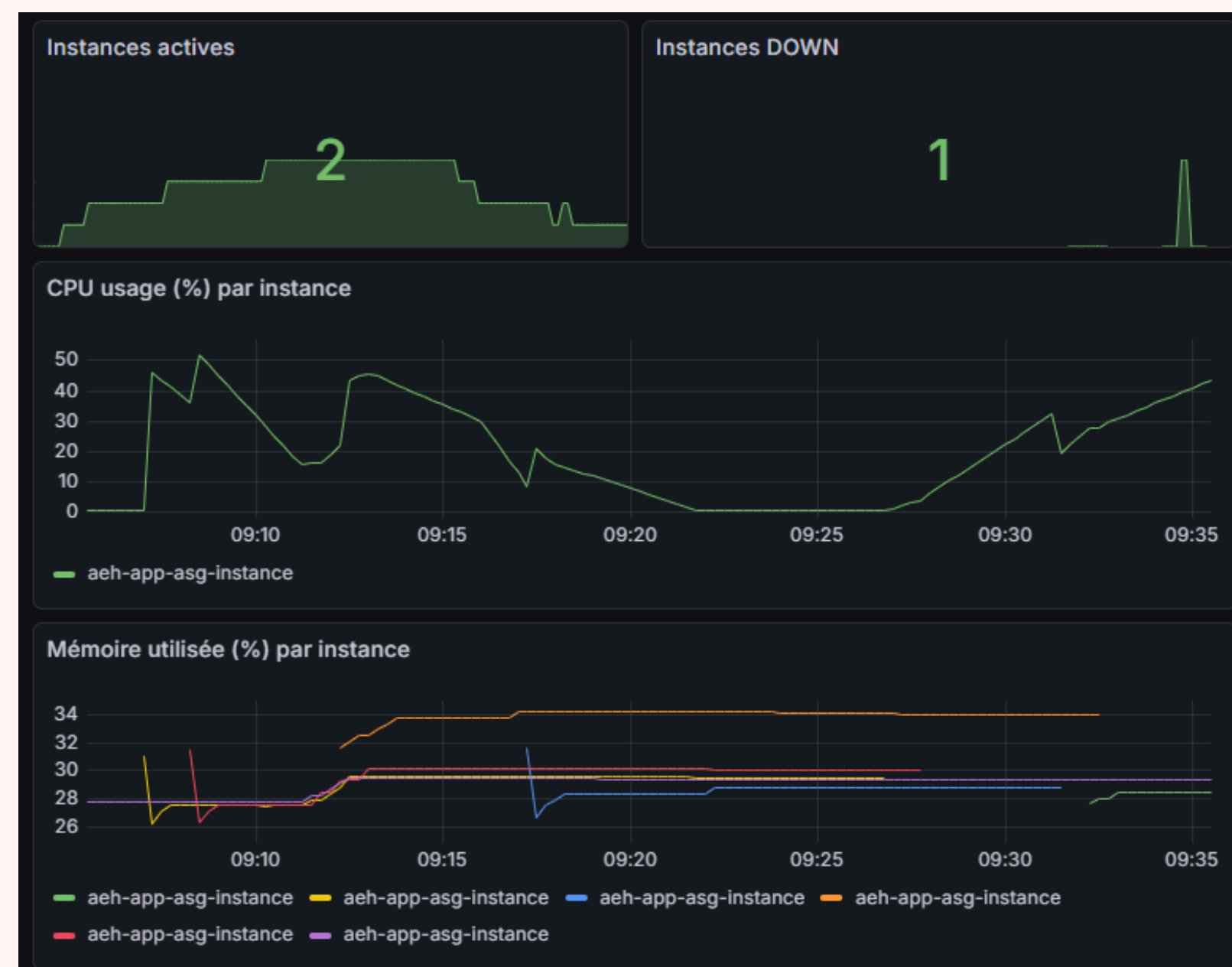
Cloudwatch + Autoscaling

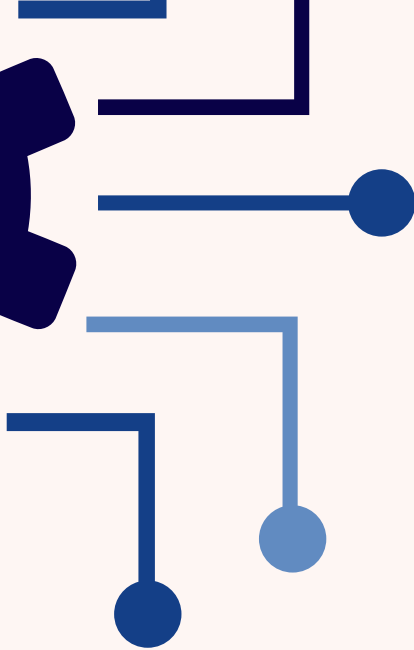
Création après 75% de CPU sur les EC2
Suppression après 30% de CPU sur les EC2

Nombre d'instance de base: 2

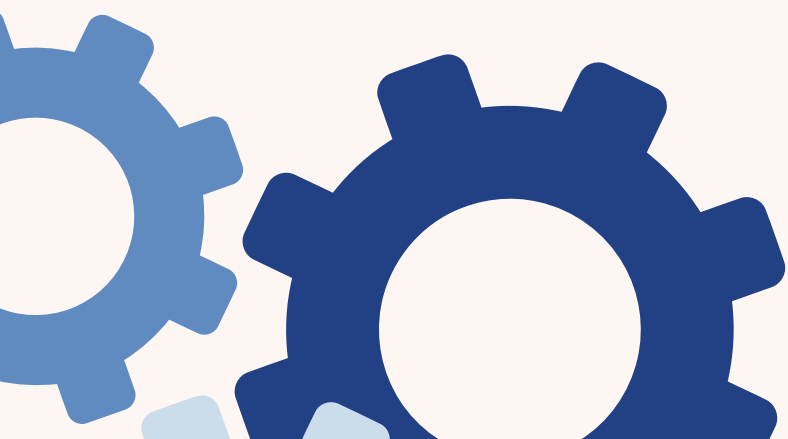
Nombre d'instance min: 2

Nombre d'instance max: 5



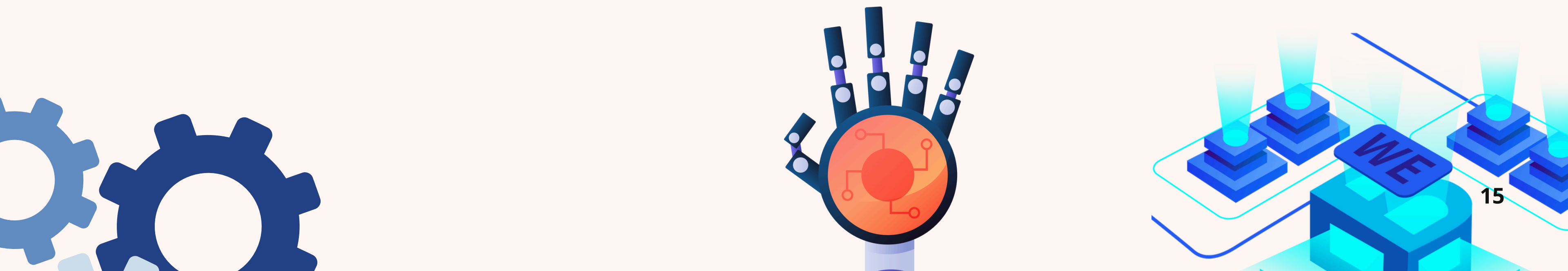


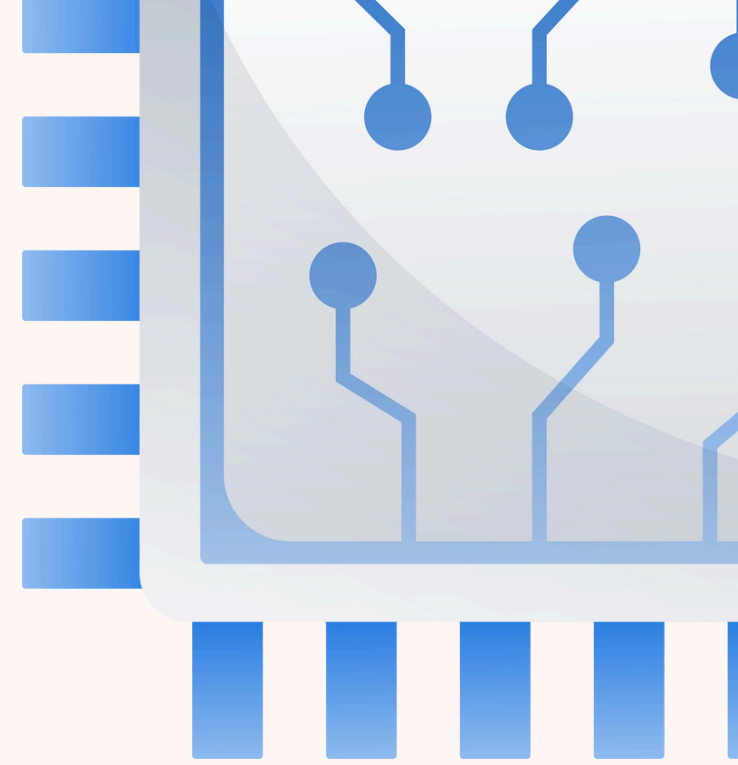
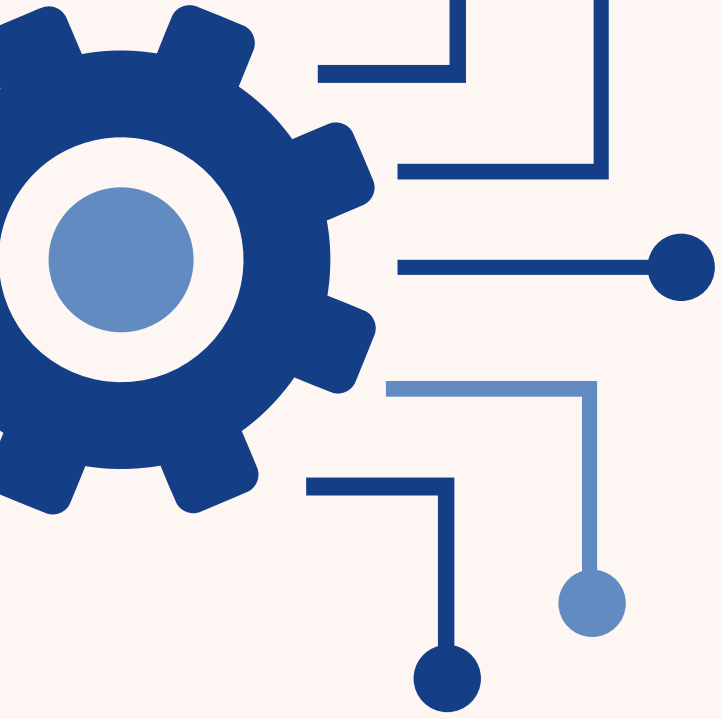
DEMO





QUESTIONS ?





MERCI

